WHEN THE TECHNOLOGIST SAYS "THE IMAGES DON'T LOOK RIGHT..."

A SYSTEMATIC APPROACH FOR MRI ENGINEERS

Use this step-by-step roadmap to investigate image quality issues from the applications perspective. Start simple → end complex.

1 PROTOCOL & CLINICAL GOAL

- Is the correct exam and sequence prescribed?
- Are the right views/planes included?
- Is contrast required? Was it given?
- Review prior studies for comparison.

Ask: What are we trying to show? Are we looking at the right thing?

2 PARAMETERS & SCAN SETTINGS

- TR / TE / TI appropriate for the goal?
- FOV, matrix, slice thickness, and bandwidth?
- Fat sat or contrast timing correct?
- Any recent protocol or parameter changes?

Ask: Are the parameters optimized for the clinical goal?

3 PATIENT POSITIONING & IMMOBILIZATION

- Is the patient centered and aligned?
- Are landmarks and prescription planes correct?
- Could motion (voluntary/involuntary) be the cause?
- Is breath-hold or gating working properly?

Ask: Is the patient in the right place and holding still?

4 COIL & HARDWARE CONNECTIONS

- Is the correct coil selected and connected?
- Any coil element failure (noise, stripes, dropouts)?
- Coil is centered and snug (no gaps or tilt)?
- Check cable connections and locking.

Ask: Is the right coil working as it should?

5 PRESCAN / CALIBRATION / CENTRING & FREQUENCY (CF)

- Were prescan/calibrations completed successfully?
- Is the FOV properly centered on the anatomy?
- Is the frequency correct (water peak)?
- Any aliasing or wrap-related problems?

Ask: Is the system measuring the right signal?

6 SHIM & B_0 HOMOGENEITY

- Poor uniformity can cause distortion, signal loss.
- Review auto shim results and map.
- Any metal, air gaps, or patient size issues?

Ask: Is the magnetic field as uniform as possible?

7 RF SYSTEM (B_1) & POWER

- Is RF gain appropriate (not too low or too high)?
- Any dielectric shading, poor fat sat, or inhomogeneity?
- Check SAR limits, duty cycle, and coil tuning/match.

Ask: Is the RF doing its job consistently?

8 GRADIENT SYSTEM

- Any streaking, geometric distortion, or ghosting?
- Check for gradient performance or heating issues.
- Review eddy current compensation if applicable.

Ask: Are the gradients performing as expected?

9 SYSTEM, SOFTWARE & WORKFLOW

- Software version, known service bulletins?
- Hardware faults or error logs?
- Network / archive / workstation issues?

Ask: Is everything else in the system healthy?

QUICK TRIAGE: MATCH THE PROBLEM TO LIKELY CAUSES

What You See	Possible Causes	Start Here
Blurry / Low Detail	Motion, wrong FOV/matrix, low SNR, thick slices	2 3 5
Signal Loss (Dark)	Wrong TI/TE, off-resonance, B_0 inhomogeneity, poor coil	2 6 4
Too Bright / Saturated	Saturation (flow/in-plane), TR too short, gain too high	2 5 7
Ghosting	Motion, flow, wrap/aliasing, phase encode problem	3 5 8
Streaking / Distortion	Metal, B_0 issues, gradient nonlinearity, eddy currents	6 8 2
Incomplete Fat Sat	B_1 inhomogeneity, wrong method, off-center	7 4 5
Venous Enhancement	Contrast timing, TR too short, flow vs tissue timing	5 2 1
Poor MRA Arteries	Saturation, slow flow, timing, suboptimal method	2 6 1

ARTIFACTS AT A GLANCE

Motion

- Blurry, double edges
- Ghosting
- Phase/Wrap

Check: 3, 5

Metal Susceptibility

- Signal void
- Distortion
- Streaking

Check: 6, 8

Chemical Shift

- Bright/dark edges
- More at high BW
- Opposite sides

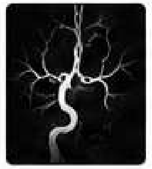
Check: 2

Truncation (Wrap)

- Anatomy appears on opposite side
- FOV too small

Check: 2, 5

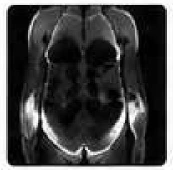
Flow / In-Plane Sat



- Loss of flow signal
- Time-of-Flight sensitive

Check: 2, 5, 1

Dielectric Shading (B1)



- Uneven fat sat
- Signal variation
- Peripheral loss

Check: 7

Gibbs (Truncation)

- Ripples near sharp edges
- High contrast boundaries

Check: 2

Nyquist / Aliasing

- Repeated anatomy
- FOV too small

Check: 2, 5

GADOLINIUM CHECKLIST

- ☑ Correct patient (name, weight, allergies, eGFR)?
- ☑ Correct dose and concentration?
- ☑ IV access patent; power injector working?
- ☑ Saline flush before and after injection?
- ☑ Timing/delay correct for sequence?
- ☑ Trigger (bolus/auto-trigger) placed correctly?
- ☑ Any adverse reaction or extravasation?
- ☑ Document lot #, dose, time, and site.

Tips

- For dynamic studies, consistency is key.
- Review prior images for timing.
- Arterial phase ≈ 15–25 sec (brain/neck) 20–30 sec (body).

SNR QUICK WINS

- Increase NEX / averages
- Increase voxel size (thickness or matrix)
- Increase bandwidth
- Use phased array coils
- Optimize parameter match for the goal
- Reduce motion

GOLDEN RULES

- ★ Quality in → Quality out.
- ★ Garbage in → Garbage out.
- ★ Change one thing at a time.
- ★ Verify before you escalate.
- ★ Document what you did.
- ★ Communicate with respect.
- ★ You are the engineer— be the problem solver.

REMEMBER

- Start simple, think systematically.
- Most image problems are applications or setup issues, not hardware failures.
- The best engineers solve problems before the patient leaves.

KEY TAKEAWAY

Follow the roadmap. Ask the right questions.
Check the basics first. Solve problems together.
Great images = Great answers = Great patient care.

★ RIGHT INFORMATION. RIGHT BALANCE. RIGHT IMAGE. Understand the sequence. Recognize the artifacts. Support the technologist. Deliver the value.